

GOOG Twitter Volume – Analytics Training and Forecast Report

- Hourly Forecasting of GOOG Twitter Volume Using SARIMAX

Executive Summary

This report analyzes hourly Twitter volume for GOOG and develops a short-term forecasting model to predict activity over the next 24 hours. A SARIMAX model with daily seasonality was used after identifying clear intraday patterns in the data. The model captures regular usage behavior effectively, while event-driven spikes remain outside the scope of this baseline approach.

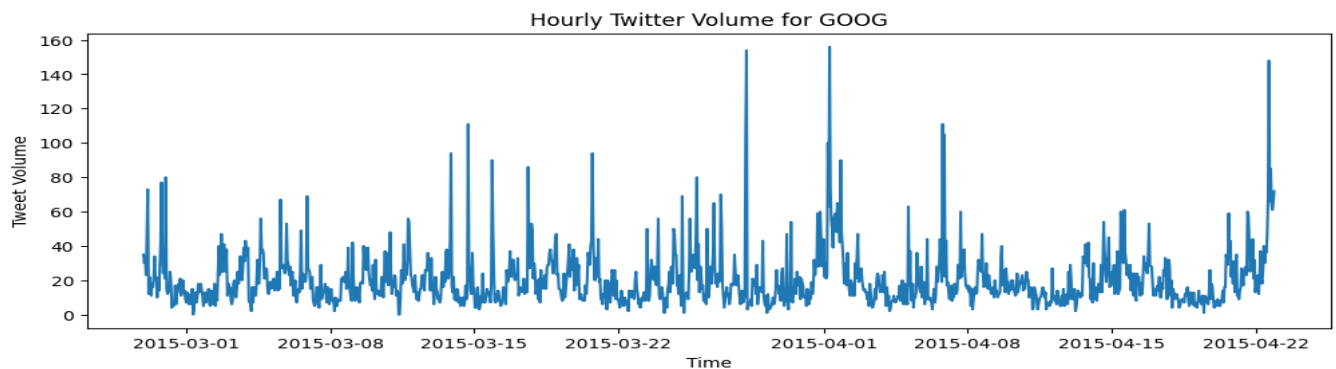
Data Overview

The dataset consists of timestamped Twitter volume data related to Google (GOOG), representing the number of tweets observed over time.

Metric	Value
Mean	20.17
Std Dev	15.93
Min	0
Max	156
1%/99%	3/83.99
Median	16
IQR Fence	-11.5/48.5

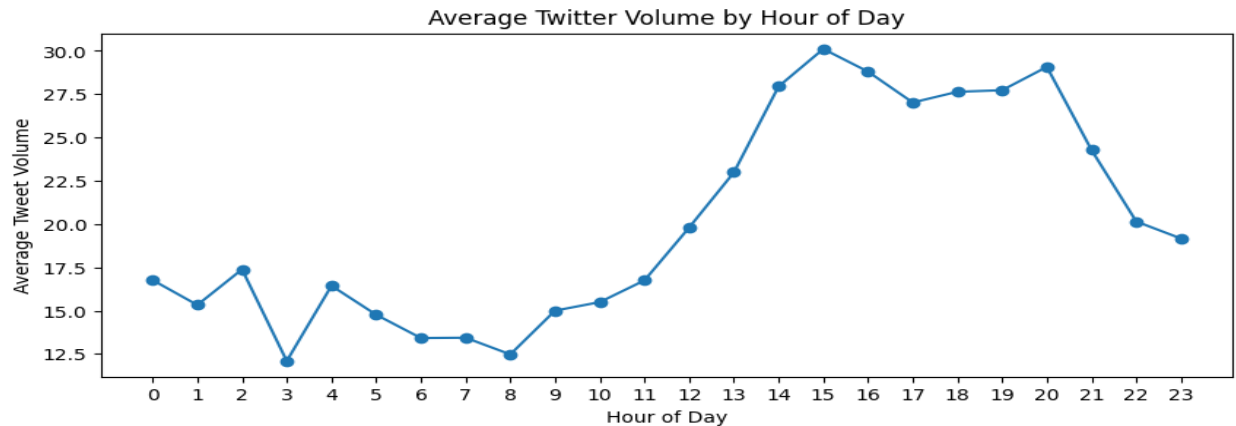
Exploratory Visualization

Plot 1: Hourly Time Series



The hourly series exhibits smoother behavior while retaining clear temporal structure. In particular, regular intraday fluctuations are visible, along with occasional sharp increases driven by external events.

Plot 2: Hour Of Day Average Volume



Analysis of average tweet volume by hour of day reveals a consistent daily usage pattern, with higher activity during daytime hours and lower activity overnight. This recurring intraday structure indicates the presence of strong daily seasonality, motivating the use of a seasonal time-series model with a 24-hour seasonal period.

Modeling Approach

- Model: **SARIMAX (Seasonal ARIMA)** applied to hourly Twitter volume
- Non-seasonal order $(p,d,q) = (3,1,2)$ to model short-term dependence and remove non-stationarity
- Seasonal order $(P,D,Q,s) = (2,1,1,24)$ to capture daily seasonality observed in exploratory analysis
- No exogenous variables included; forecasts are driven solely by historical tweet volume
- Model evaluated on a 24-hour holdout window and refit on the full dataset for final forecasting

Model Evaluation and Results

Evaluation:

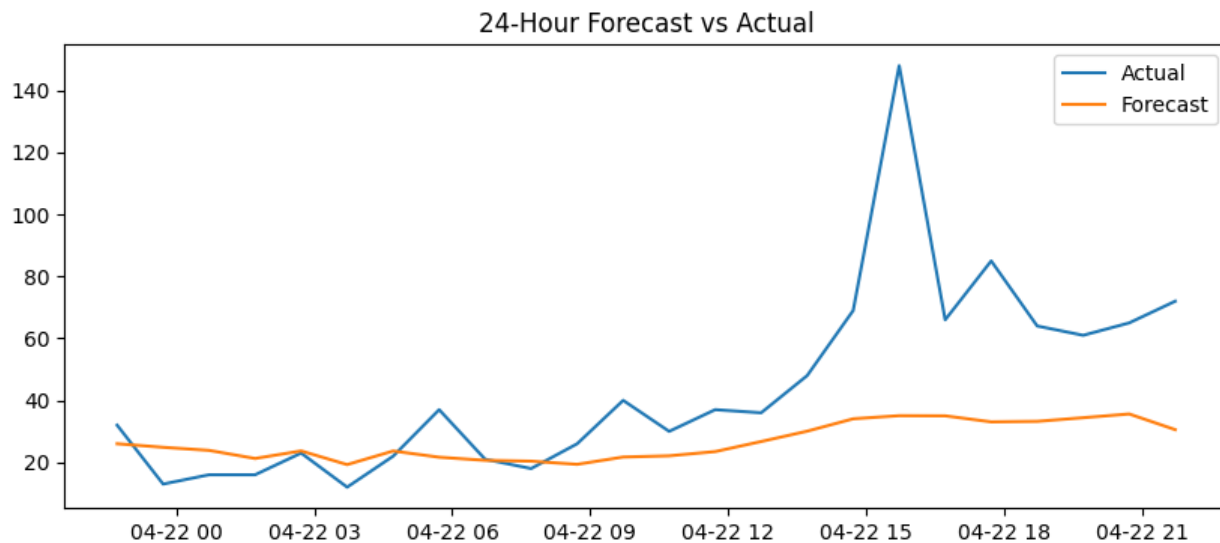
- Validation performed using a **time-based 24-hour holdout**
- Model trained on historical hourly data excluding the final 24 hours
- Forecasts generated strictly on unseen data

Performance Metric:

<i>Metrix</i>	<i>Value</i>
<i>Mae</i>	<i>20.5 tweets/hour</i>
<i>Rmse</i>	<i>31.3 tweets/hour</i>
<i>Mape</i>	<i>26.6%</i>

Forecast Accuracy:

- Forecast is smooth stable across the horizon
- Unable to capture event that leads to spikes
- Model does capture overall level and daily pattern



Forecast Generation

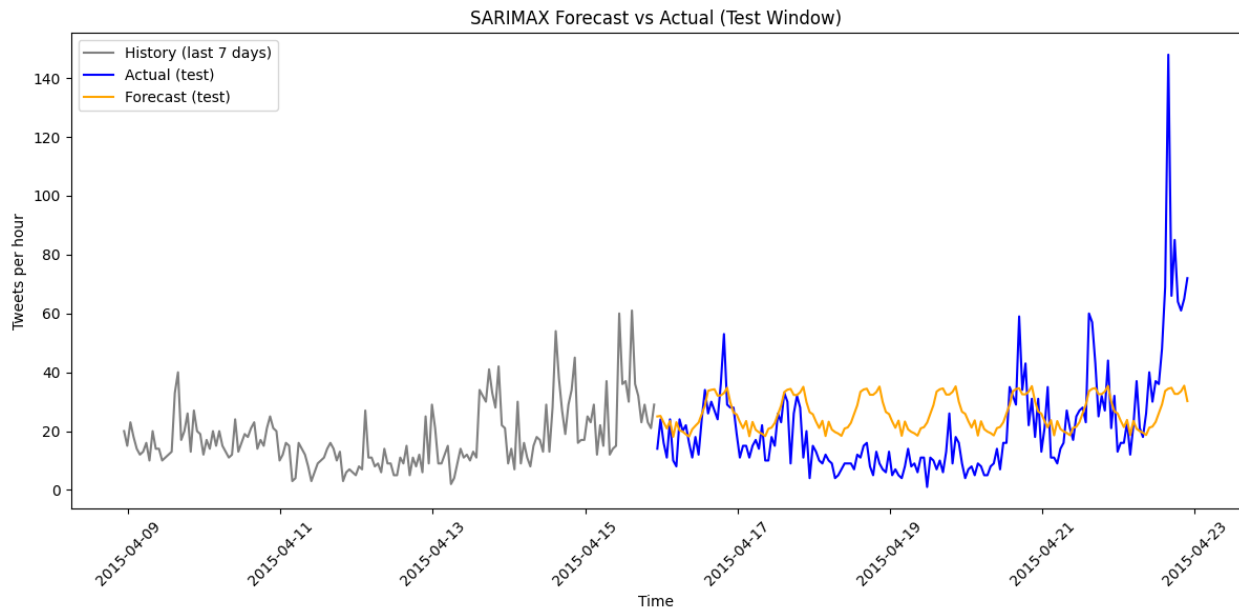
- Model refit on all available hourly data after evaluation
- Same SARIMAX configuration retained to ensure consistency
- Forecast horizon: next 24 hours

Forecast Output

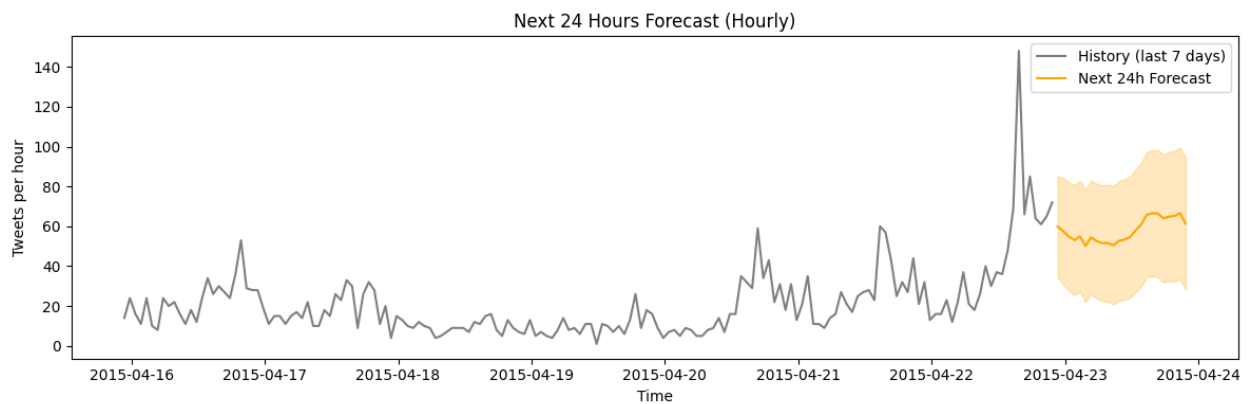
- Point forecasts generated for each hour
- 95% confidence intervals included to quantify uncertainty

- Forecast reflects expected daily usage pattern based on historical behavior

Final Test Forecast:



Plot next 24 Hour Forecast:



Limitation:

- Model does not incorporate external drivers such as news or Market events
- Event driven spikes cannot be anticipated with univariant approach
- Future improvements may include:
 - Exogenous variables (events, market hours)
 - Recent-window retraining or alternative forecasting model for comparison